Karim Zaghw

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Research Profile

I am interested in how learning agents construct useful representations from raw experience: discovering concepts and causal structure through interaction, choosing informative actions, and revising their models when evidence or goals change. My current work draws on causal representation learning, reinforcement learning, and Bayesian methods, with an emphasis on identifiability, abstraction, and decision-making under uncertainty.

Education

MSc Computational Neuroscience 2024/10–Present
University of Tübingen Tübingen, Germany
Expected graduation: 07/2027; current grade: 1.0 on the German scale (1.0 best).

Bachelor of Electrical Engineering 2016/09–2021/05
Cairo University Cairo, Egypt
Major: Communication and Computer Engineering; GPA: 3.74/4.00; Distinction with Honours.

Selected Publications

Karim Zaghw, Peter Dayan, and Georgy Antonov (2026). *Weight a Moment: Risk-Aware Exploration with Bayes. Reinforcement Learning Journal*, vol. 7; paper.

Karim Zaghw, Andrew Pashea, Marc Pritsch, Wouter Nuijten, Karl Friston, and Lancelot Da Costa (2026). *Renormalising Generative Models for Active Inference: Foundations, Derivations, and Verification.* Accepted at IWAI 2026; Springer CCIS proceedings forthcoming; arXiv.

Karim Zaghw, Wieland Brendel, and Patrik Reizinger (2026). *When Do Learned Blocks Preserve Causal Meaning? Block Identifiability and Constructive Causal Abstractions.* Under review; arXiv:2610.18155.

Research Presentations

Weight a Moment: Risk-Aware Exploration with Bayes — contributed talk and poster, RLC, Montréal, Canada 2026

Research Experience

Student Research Assistant 2025/01–Present
Department of Computational Neuroscience, Max Planck Institute for Biological Cybernetics Tübingen, Germany

- Developed Bayesian Moment Learning, a model-free distributional reinforcement-learning method that represents epistemic uncertainty over risk-relevant moments and extends value-of-perfect-information exploration to moment-based risk objectives; published in the Reinforcement Learning Journal and presented at RLC 2026.
- Current project: developing posterior-probability and value-of-computation stopping rules for Bayesian Monte Carlo tree search while studying uncertainty propagation under non-stationary backups.

MSc Research Rotation: Causal Abstraction for Overcomplete Representations 2026/07/01–2026/09/23
Robust Machine Learning Group, ELLIS Institute Tübingen Tübingen, Germany

- Characterized when block-identifiable representations preserve high-level causal abstractions under reparameterization, with a maximality result and synthetic intervention-transfer experiments.

MSc Research Rotation: Renormalising Generative Models 2026/04/15–2026/06/16
Methods of Machine Learning Group, University of Tübingen Tübingen, Germany

- Reconstructed and verified a hierarchical active-inference framework, translating core MATLAB/SPM routines into Python and clarifying its inference, planning, and structure-learning derivations; work accepted at IWAI 2026.

Teaching and Mentorship

Internship Mentor, Raisa Energy 2023/07–2023/10
Mentored a Text-to-SQL chatbot project using LLMs; achieved over 90% accuracy on daily queries; project write-up.

Junior Teaching Assistant, Operating Systems, Cairo University 2021/01–2021/06
Assisted in teaching, writing, and grading assignments, projects, and quizzes for 100+ students.

Additional Training

MITx MicroMasters Program in Statistics and Data Science 2023/02–2024/04
MITx (MIT Open Learning), edX; SDS General Track.

Deep Learning Specialization, DeepLearning.AI 2022/12–2023/05

Professional Experience

Data Scientist / Data Engineer, Raisa Energy 2021/07–2024/09
Built forecasting, risk-modeling, and data systems across 20+ projects tracking $1B+ in assets.